Trackerr – Components Map

A weekly mutual-fund check-up for Meera, in plain words. **All numbers are worked out by ordinary code. The AI (Google Gemini Flash only) just explains them** – and its answer is checked before anyone sees it.

Live site: **https://trackerr-amlan-mahapatra-1955.vercel.app** Code: **https://github.com/amlanmahapatra-bit/meera-mf-tracker**

EVERY DAY · 7:00 AM IST
Saves the latest fund prices and index levels

EVERY SATURDAY · 7:30 AM IST
Builds that week's report and emails it with a PDF

SUN–TUE · SAFETY NET
If Saturday's email was missed, builds and sends it again (never twice)

ANY TIME · BY HAND
"Update now" button; Meera adds or removes a fund (passcode, no code changes)

1. Get live data
from free public sources

mfapi.in
Daily price (NAV) of every fund, full history. Also used to search funds by name.

Yahoo Finance
Nifty 50, Nifty Midcap 150, Nifty Smallcap 250, Nasdaq 100 and the dollar–rupee rate.

GNews
This week's India, economy and world market headlines.
Backup: Google News RSS, switches on automatically

2. Save it
Supabase database

Prices & index history
Every fund price and index close, saved so the site still works if a source is down.

Your 5 funds
Which funds Meera tracks, with their market index and similar funds. Add or remove any time.

Weekly snapshots & reports
Every Saturday saved – 5 years of weeks – for the trend chart.

Run log & email log
Every job's start, end and result; every email sent. Jobs are safe to repeat – nothing is duplicated.
Only our server can read it (no public access)

3. Do the maths
plain code – no AI, fully tested

Returns
Week, month, year-so-far, and 1 day to 5 years. Weekends and holidays use the last price before.

Compare
Fund vs its market index and vs the average of similar funds, on the same dates.

"Market or fund?" split
How much of the week's move came from the whole market, and how much from the fund itself.

Traffic light & verdict
Green / amber / red per fund. Overall: "No action needed", "One fund worth watching" or "Consider reviewing your allocation", with a reason.

4. Explain in plain words
Gemini Flash via OpenRouter – the only AI

Gemini writes the words
3 short lines per fund (what happened, how it compares, why – from a real headline), a longer story, and a market summary.

Checker
Rejects invented numbers, jargon, buy/sell advice and made-up headlines. Up to 3 tries.

Template fallback
If Gemini is down or keeps failing, code writes the same report from the numbers. A report always exists.

5. Deliver to Meera
what she actually sees

Website (Vercel)
Dashboard with verdict and 5 traffic-light cards · fund pages with 5-year price charts · "How it works".

Weekly report page
The whole week on one printable page.

Saturday email + PDF (Resend)
The same detailed report, with a PDF attached.

"Ask about my money" chat
Answers her questions using only her saved report and the week's headlines.

Add / remove a fund form
Search by name or paste a scheme code. Passcode protected.

Built so it never breaks – and so the numbers can be trusted

- ✔ Source down? Uses the saved copy and says "data as of" – never a blank page
- ✔ News source down? Switches to Google News automatically
- ✔ AI down or wrong? Checker + template, so the report still goes out
- ✔ Email failed? Re-sent Sun–Tue, and never sent twice
- ✔ Every job is logged with its status (visible at /api/health)
- ✔ Secrets live only in Vercel settings; timers need a secret, fund changes need a passcode

Colour key: Outside data · Database · Our code · AI (Gemini Flash) · What Meera sees

Built with Next.js, TypeScript, Supabase and Vercel · Honest limits: no free benchmark exists for the debt fund (compared with similar funds only) and Nifty Smallcap 100 has no free data (Smallcap 250 is used).